Tutorial 7 — Solutions

Question 1 Reading Club Ratings

The rating matrix A (rows = books, columns = students, in the order Rohit, Aisha, Kalp, Yogesh) and the weight vector w are

$$A = \begin{pmatrix} 4 & 3 & 5 & 2 \\ 2 & 5 & 3 & 4 \\ 3 & 4 & 2 & 5 \\ 5 & 2 & 4 & 3 \end{pmatrix}, \qquad w = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 2 \end{pmatrix}.$$

(a) Dimensions. A is a 4×4 matrix (4 books $\times$ 4 students), and w is a 4×1 column vector (one weight per student).

(b) Is Aw defined? The product Aw is defined whenever the number of columns of A equals the number of rows of w. Here A has 4 columns and w has 4 rows, so

$$A_{(4\times4)}\, w_{(4\times1)} \longrightarrow s_{(4\times1)},$$

and the product is mathematically defined, producing a 4×1 vector.

(c) Compute $s = Aw$.

$$s = \begin{pmatrix} 4 & 3 & 5 & 2 \\ 2 & 5 & 3 & 4 \\ 3 & 4 & 2 & 5 \\ 5 & 2 & 4 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 4(2)+3(1)+5(3)+2(2) \\ 2(2)+5(1)+3(3)+4(2) \\ 3(2)+4(1)+2(3)+5(2) \\ 5(2)+2(1)+4(3)+3(2) \end{pmatrix} = \begin{pmatrix} 8+3+15+4 \\ 4+5+9+8 \\ 6+4+6+10 \\ 10+2+12+6 \end{pmatrix} = \begin{pmatrix} 30 \\ 26 \\ 26 \\ 30 \end{pmatrix}.$$

So $s_1 = 30$ (Feynman Lectures), $s_2 = 26$ (The Boy Who Loved), $s_3 = 26$ (Pride and Prejudice), $s_4 = 30$ (A Brief History of Time).

(d) Highest weighted score. The maximum entry of s is 30, attained by both *The Feynman Lectures on Physics* and *A Brief History of Time*. These two books are tied for the highest overall weighted score.

Question 2 Spanning and Orthogonality

$v_1 = (1, 2, -1)$, $v_2 = (2, -1, 0) \in \mathbb{R}^3$.

(a) Can v_1, v_2 span $\mathbb{R}^3$? Spanning $\mathbb{R}^3$ requires at least 3 linearly independent vectors, since $\dim(\mathbb{R}^3) = 3$. Here we are given only 2 vectors, so at most a 2-dimensional subspace (a plane through the origin) can be spanned. Hence v_1 and v_2 cannot span all of $\mathbb{R}^3$, regardless of whether they are independent.

(b) Orthogonality.

$$v_1 \cdot v_2 = (1)(2) + (2)(-1) + (-1)(0) = 2 - 2 + 0 = 0.$$

Since the dot product is 0, the vectors v_1 and v_2 are orthogonal.

Question 3 Rotation Matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta = 90^\circ, \qquad v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

(a) Numerical matrix R. With $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(b) Transformed vector $v' = Rv$.

$$v' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0(2) + (-1)(1) \\ 1(2) + 0(1) \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

(c) Geometric picture. $v = (2, 1)$ lies in the first quadrant; $v' = (-1, 2)$ lies in the second quadrant. Plotting both from the origin shows that v' is obtained by rotating v through 90° in the counter-clockwise (anticlockwise) direction, consistent with the standard convention for $R(\theta)$.

(d) Transpose and $R^\top R$.

$$R^\top = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad R^\top R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

So indeed $R^\top R = I$.

(e) Is R orthogonal? Since $R^\top R = I$ (equivalently $R^\top = R^{-1}$), the matrix R is orthogonal. This holds for any rotation matrix $R(\theta)$.

(f) Lengths of v and v'.

$$\|v\| = \sqrt{2^2 + 1^2} = \sqrt{5}, \qquad \|v'\| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}.$$

The lengths are equal: $\|v\| = \|v'\|$. This illustrates that a rotation matrix is norm-preserving (an isometry) — rotations change direction but never the magnitude of a vector.

Question 4 Rotation Followed by Scaling

Point $P(2, 3)$ is rotated 45° anticlockwise, then scaled by $S = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

Rotation matrix at 45°:

$$R(45^\circ) = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

Combined transformation matrix (scale after rotate, so $M = SR$):

$$M = SR = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \frac{3\sqrt{2}}{2} & \frac{3\sqrt{2}}{2} \end{pmatrix}.$$

Applying it to P:

$$P' = M\begin{pmatrix}2\\3\end{pmatrix} = \begin{pmatrix}\sqrt{2}(2) - \sqrt{2}(3)\\ \frac{3\sqrt{2}}{2}(2) + \frac{3\sqrt{2}}{2}(3)\end{pmatrix} = \begin{pmatrix}-\sqrt{2}\\ \frac{15\sqrt{2}}{2}\end{pmatrix}.$$

So the final coordinates are

$$P' = \left(-\sqrt{2},\ \tfrac{15\sqrt{2}}{2}\right) \approx (-1.414,\ 10.607).$$

Question 5 Shear, Rotation, and Scaling

Point $P(1, 2)$ undergoes, in order: (1) x-shear with $k = 2$, (2) $90°$ anticlockwise rotation, (3) uniform scaling by 2.

Individual matrices:

$$\text{Shear: } Sh = \begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix}, \qquad \text{Rotation: } R = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}, \qquad \text{Scaling: } Sc = \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}.$$

Combined transformation matrix. Since the shear is applied first and scaling last, the matrices compose in reverse order:

$$M = Sc\,R\,Sh.$$

First,

$$R\,Sh = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\ 0 & 1\end{pmatrix} = \begin{pmatrix}0 & -1\\ 1 & 2\end{pmatrix}.$$

Then,

$$M = Sc\,(R\,Sh) = \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 2\end{pmatrix} = \begin{pmatrix}0 & -2\\ 2 & 4\end{pmatrix}.$$

Final coordinates.

$$P' = M\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}0 & -2\\ 2 & 4\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}0(1) + (-2)(2)\\ 2(1) + 4(2)\end{pmatrix} = \begin{pmatrix}-4\\10\end{pmatrix}.$$

Check step by step: Shear gives $(1, 2) \to (1 + 2 \cdot 2, 2) = (5, 2)$; rotating gives $(5, 2) \to (-2, 5)$; scaling gives $(-2, 5) \to (-4, 10)$ — consistent with M.

So the combined matrix is $M = \begin{pmatrix}0 & -2\\ 2 & 4\end{pmatrix}$ and the final point is $P'(-4, 10)$.